

Boosting proton beam therapy potential

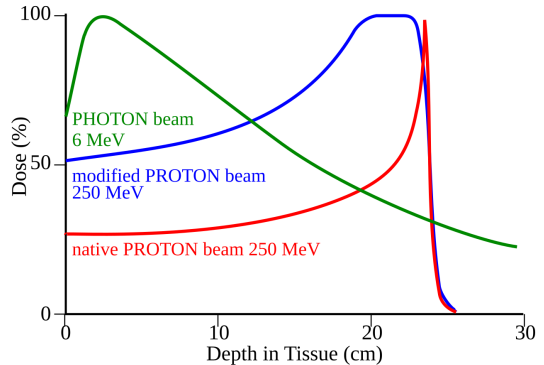
Clinical motivation and measurement principles

DIPC CRYSP group

Working draft — September 14, 2026

What is proton therapy?

- A form of **external-beam radiotherapy** that uses accelerated protons to destroy tumour cells.
- Protons are **positively charged particles**: their energy determines how deep they travel in tissue.
- Their dose peaks near the end of their range (the **Bragg peak**), then falls sharply.
- Combining different proton energies covers the tumour volume, while **reducing dose beyond it** compared with X-rays.



Schematic depth–dose profiles: a single proton energy, a spread-out peak and an X-ray beam.

Image: D. Ilyin / A. A. Miller, Wikimedia Commons (CC0).

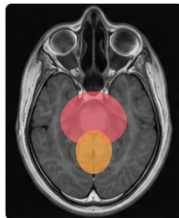
Reference: National Cancer Institute, External Beam Radiation Therapy.

Example: Brain tumours

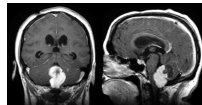
- Brain and CNS tumours are the **most common solid tumours in children**: about **300 new cases a year in Spain**.
- In adults, malignant brain/CNS tumours account for roughly **4,400 cases a year in Spain** (about 1.5% of cancers).
- Childhood ependymomas often arise near the **cerebellum and brainstem**. The goal is to cover the tumour while sparing these structures.

Childhood estimate: Pediatría Integral (2021), based on RETI.

Adult figure: approximate, from REDECAN 2026 all-age malignant CNS estimate.

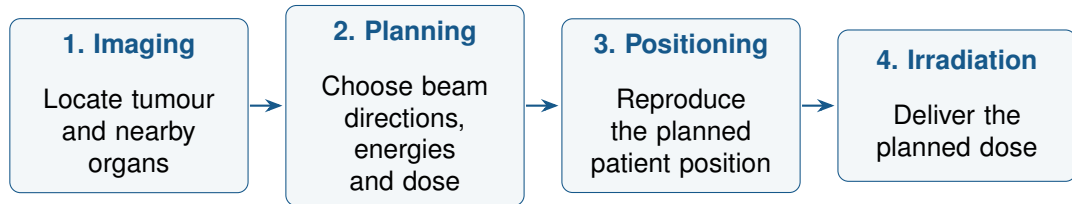


Brain/skull-base example: tumour (pink), brainstem (orange).



Fourth-ventricle ependymoma (MRI).

From treatment planning to delivery



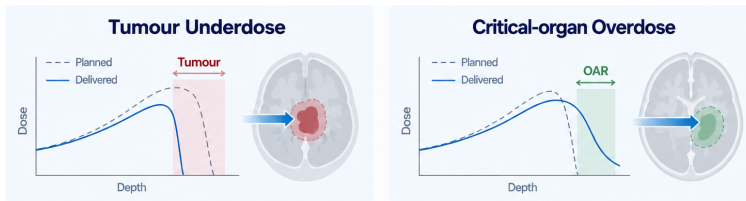
Delivered dose can differ from plan

- Uncertainty in tissue stopping power.
- Anatomical changes, positioning and motion.

Unlike photons, protons are absorbed in the patient's body

The plan delivers a **prediction** of proton range. **How to check it?**

Why range uncertainty matters



- Systematic errors in proton range **estimation** are of the order of **3.5%**.
- The required planning margins are of the order of several mm.
- Conservative beam choices protect nearby organs, but can limit coverage when tumour and organ are close.

Example at 10 cm depth

$$0.035 \times 100 \text{ mm} + 1 \text{ mm} \\ = 4.5 \text{ mm}$$

3.5% range allowance plus 1 mm.
Margins depend on the centre and treatment site.

Range uncertainty: Paganetti, Phys. Med. Biol. 57, R99 (2012). OAR = organ at risk.

What could better range verification change?

- With less range uncertainty, planners can reduce dose to **healthy tissue near the tumour**.
- They may also use **beam directions otherwise avoided** because a critical organ lies near the stopping point.
- Verification must show that the uncertainty is sufficiently small before these changes can be adopted.

A planning study: 27 brain-tumour patients

Range uncertainty reduced from **3% to 2%**.

24 of 27 patients (89%) had a relevant dose reduction to at least one organ at risk.

In brain tumours, proton range uncertainties of **1 mm or less** can in principle be achieved.*

Taasti et al., British Journal of Radiology 96, 20230110 (2023).

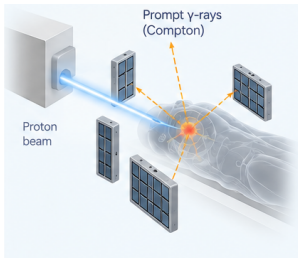
Beam-direction benefit: Tattenberg et al., Medical Physics (2022).

* J.J. Gómez-Cadenas et al., *A High-Sensitivity In-Room CRYSP Scanner for Distal-Edge Verification in Proton Therapy*, paper in preparation, available upon request.

Complementary imaging for proton therapy

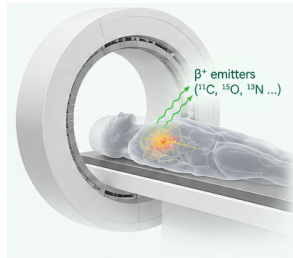
Compton camera

In-beam: prompt γ rays



PET

In-room, post-treatment: β^+ activity



R&D studies support Compton cameras and PET for **proton range verification**. No dedicated commercial solution identified.

Safavi et al., Nat. Commun. 17, 206 (2026).

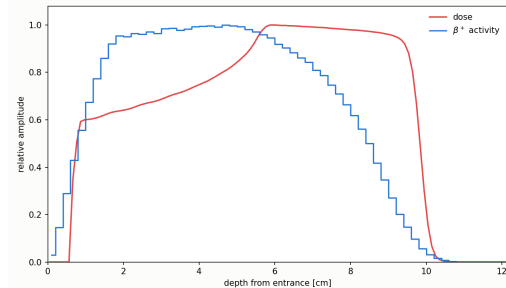
Sharma et al., Med. Phys. 52, 1123 (2025).

Parodi et al., Eur. J. Nucl. Med. Mol. Imaging 51, 183 (2024).

Zhu et al., Phys. Med. Biol. 56, 6425 (2011).

PET: an *in vivo* probe of where the beam stopped

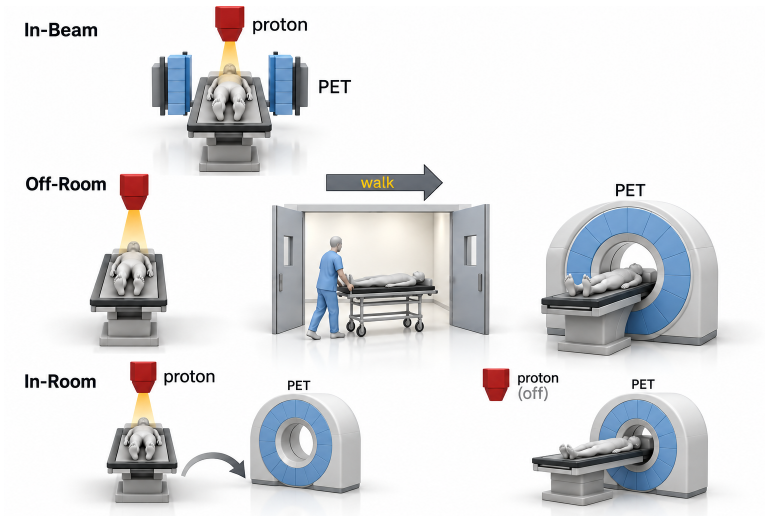
- Above the nuclear-reaction threshold, protons create β^+ emitters along their path — mainly ^{15}O and ^{11}C .
- Their annihilation γ 's are imaged in coincidence: a PET image of the induced activity.
- The activity falloff sits just upstream of the dose edge; its distal edge tracks the range.



Illustrative dose and β^+ activity profiles.

The activity edge is compared with a prediction to estimate the range.

PET range verification: offline, in-beam and in-room



PET range verification: offline, in-beam and in-room

Offline

patient moved to
a nearby PET/CT

15–30 min later

washout, ^{15}O gone,
co-registration

In-beam

two planar heads
at the treatment
position

~ zero delay

limited angular
coverage

In-room

full ring, next
to the treatment
room

~ 2 min delay

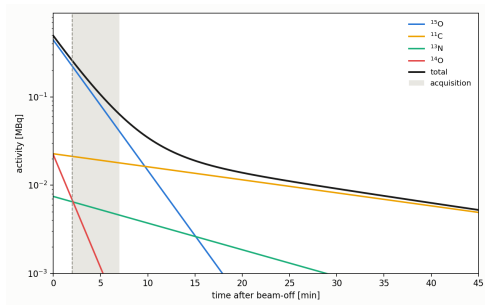
full coverage,
high sensitivity

The trade-off is **delay** vs. **angular coverage** / **scanner quality**.

Offline results in large uncertainties

A delay of 15–30 minutes costs on three fronts:

- **Sensitivity**: the abundant ^{15}O (2 min) has vanished — only the sparse ^{11}C tail remains.
- **Biological washout**: perfusion clears the emitters; hard to model, patient-dependent.
- **Co-registration**: the patient is set up afresh on a separate scanner — a few-mm misalignment maps straight into an apparent range shift.



^{15}O dominates the first minutes, then decays; ^{11}C carries the late, weak signal.

Online/In-room tradeoffs

In-beam (short delay, same bed)

- Retains the short-lived ^{15}O , essentially no delay.
- But the beam must pass: only **two planar heads** fit.
- Reduced angular coverage $\Rightarrow$ **low sensitivity** and limited-angle image artefacts.

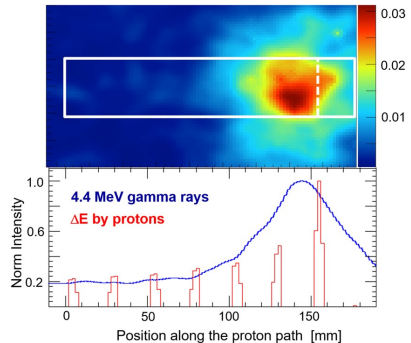
In-room (just after, full ring)

- A **full closed ring** $\Rightarrow$ full coverage, high sensitivity, clean image.
- Same couch $\Rightarrow$ a known rigid transform and a short transfer, without separate-scanner repositioning.
- The ~ 2 -min delay is **bought back** by sensitivity and a **better scanner**.

A **high-sensitivity in-room ring** combines short transfer time with complete angular coverage and high sensitivity.

Prompt gammas: tracking the proton range during irradiation

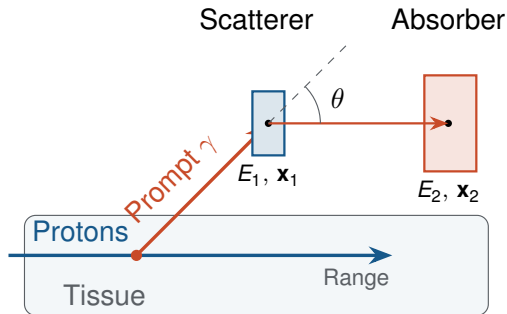
- Nuclear reactions along the proton path leave excited nuclei in the tissue. Their de-excitation emits **prompt γ rays** with energies of several MeV.
- Emission occurs **during irradiation**. Its distal falloff tracks the proton range; the relation depends on tissue composition and the selected γ energies.
- The measured **4.4 MeV γ profile** shown here concentrates near the Bragg peak and follows the proton range.



Measured prompt- γ profile and simulated proton energy deposition. 70 MeV protons; spaced PMMA slabs.

Imaging prompt gammas with a Compton camera

- A **Compton camera** measures the positions and energies of successive interactions of the **same γ ray**.
- The photon scatters in the first detector and is absorbed in the second. The deposited energies determine the **Compton scattering angle**.
- The two interaction positions define the cone axis; the scattering angle defines its opening. The emission point lies on this **cone of possible origins**.
- Combining many events reconstructs the **emission profile and its distal edge**.



One photon, two detector interactions: Compton scattering followed by absorption.

Hueso-González et al., Front. Oncol. 6:80 (2016).

Hybrid in-beam PET and Compton imaging (2022).

PET and Compton imaging: complementary measurements

	Prompt gammas / Compton	Induced activity / PET
Signal	Single prompt γ rays, several MeV	Pairs of 511 keV annihilation photons
Timing	During beam delivery	In the proposed in-room scheme, shortly after irradiation
Information	Emission profile along the beam; comparison with expected range	Activity distribution; comparison with expected distal edge
Main limitation	Few useful multi-interaction events and beam-related background	Decay, biological washout and patient registration

DIPC proposed strategy: Compton imaging for monitoring during irradiation, followed by an in-room PET measurement within minutes of beam stop. Combine both results to minimize systematic errors.

CRYSP: PET with cryogenic cesium iodide

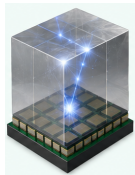
Pure Csl at ~ 100 K



$\sim 10^5$ photons/MeV.

Abundant material; lower crystal cost than rare-earth scintillators.

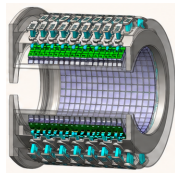
Monolithic 3D detection



Large crystals read out by silicon photomultipliers (SiPMs).

Millimetre-scale 3D resolution in simulation; measured energy resolution below 7% at 511 keV.

Full-ring PET

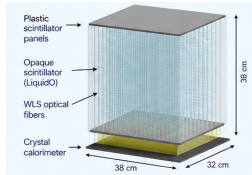


Total-body PET design based on these crystals.

High sensitivity at lower detector cost is the design goal.

COCOA: a compact Compton camera for MeV gamma rays

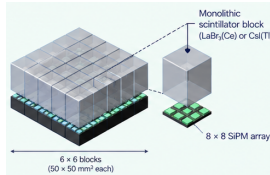
LiquidO scatterer



Opaque liquid scintillator with a 3D grid of wavelength-shifting fibres.

Reconstructs the **position and energy** of the first interaction.

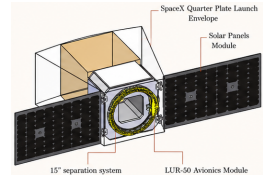
Crystal absorber



Monolithic scintillator blocks read out by SiPM arrays.

Measures the scattered photon: **3D position and energy**, including depth of interaction.

Space application

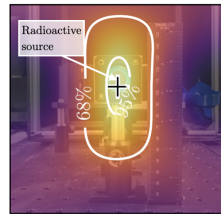
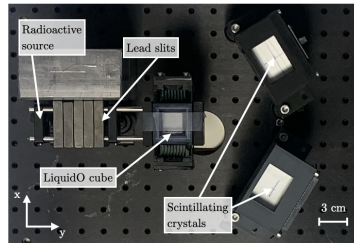


Designed for **MeV gamma-ray astronomy**.

Compact geometry for microsatellites or high-altitude balloons; proposed LUR-50 integration shown.

COCOA: demonstrated scalability and efficiency

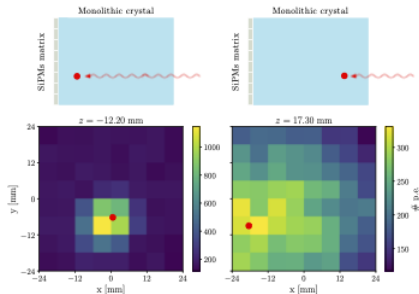
- **Scalability**: the scatterer is an opaque liquid scintillator read by a lattice of WLS fibres, so the sensitive volume scales with the cost of liquid and fibres, not with crystal growth. Inorganic crystals and CZT are limited to cm-scale units that must be tiled at high cost.
- **Efficiency**: stochastic light confinement resolves multiple Compton vertices inside a single monolithic volume, allowing a thick, large-acceptance scatterer.
- Gamma-ray imaging has already been **demonstrated** in a benchtop experiment.



Benchtop setup and reconstructed ^{60}Co image.

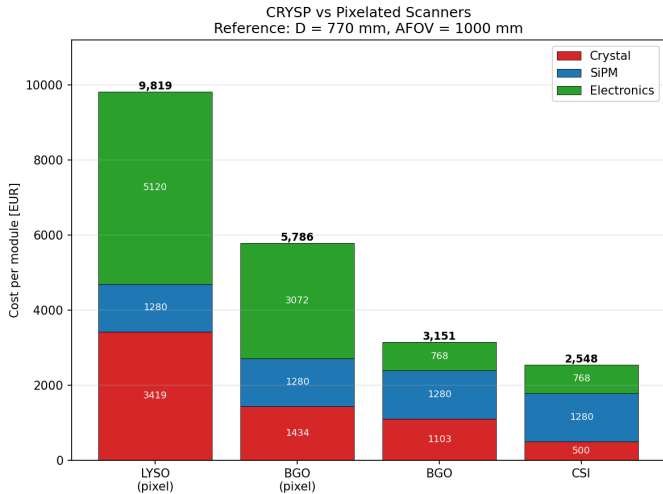
CRYSP: *CRYogenic Sensitive PET*

- Scintillators **CsI** and **BGO** gain dramatically in light yield when cooled — at 77 K, CsI reaches ~ 100 photons/keV and BGO ~ 29 photons/keV.
- In both cases, high yields translate into excellent energy resolution (6% CsI, 10% BGO), which in turn is essential for efficient scatter rejection.
- Monolithic crystals + SiPMs permit 3D position reconstruction using neural networks with high accuracy and no parallax error.

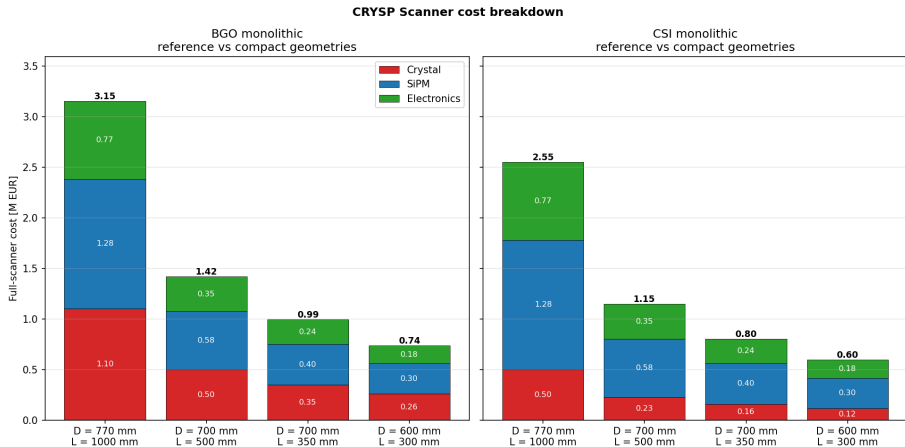


3D reconstruction in a monolithic crystal.

Scanner cost breakdown



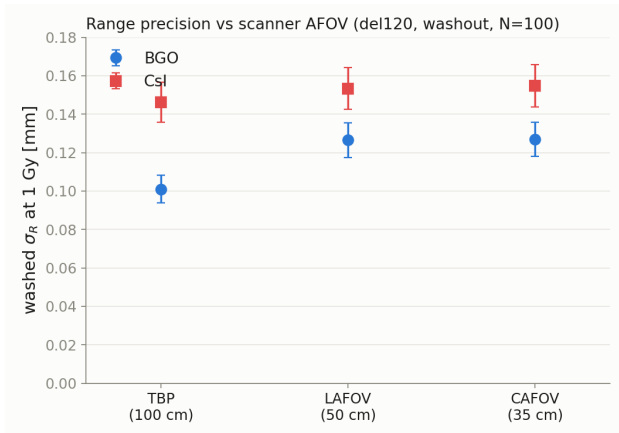
Total scanner cost



ARGOS baseline: BGO, 600 mm bore, 300 mm length, 0.74 M€ in crystal, SiPMs and electronics. Backup: Csl(Tl) at room temperature, 0.60 M€.

Reference protocol: all scanner configurations are sub-mm

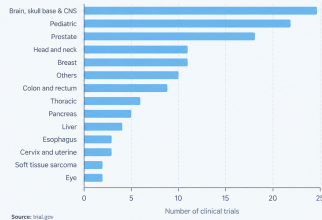
- **Protocol:** 1 Gy, 300 s acquisition, starting 120 s after beam-off; washout included.
- BGO: 0.101–0.127 mm; Csl: 0.146–0.155 mm.
- BGO is more precise through higher sensitivity; Csl remains a competitive, lower-cost option.
- These values are the **statistical precision of the fitted activity edge**.



Finite-pool-corrected σ_R for the reference protocol ($N = 100$).

Clinical evidence for proton therapy

Increasing clinical evidence for indication



Majority in CNS, pediatric, prostate, breast, and lung

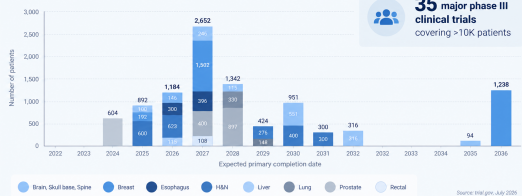


Mainly focused on side-effects reduction

Expanding clinical evidence pipeline

Supporting momentum for proton therapy

Number of patients per ongoing phase III clinical trial (proton therapy vs. radiotherapy)



35 major phase III clinical trials covering >10K patients

Phase III trials are building the **clinical evidence** for wider use of proton therapy, particularly to **reduce side effects**.

Supplied charts cite trial.gov (phase III chart: July 2026). Trial counts and projected completion dates require registry verification.

Boosting proton therapy

1. Proton beam dose monitoring **in real time** is a real need that will become more pressing as the number of centres grows and the number of treated patients increases.
2. R&D on monitoring techniques has been going on for about 20 years. PET and gamma cameras offer the potential for **day-by-day, case-by-case monitoring** and distal-edge measurements that could reduce underdose while minimising the risk of overdose.
3. However, these systems have **not yet become standard medical practice**.

Clinical status: Current status and upcoming developments for online adaptive proton therapy, Front. Oncol. (2025).

The DIPC group

The DIPC group is a leading neutrino physics group in Spain, with extensive expertise in **nuclear instrumentation**.

Scientific leadership and concepts

- Led by **J.J. Gomez-Cadenas**: co-recipient of the **2016 Breakthrough Prize in Fundamental Physics** as a member of K2K and T2K, with group leadership roles in neutrino physics.
- Two ERC grants: **AdG-2013** and **SyG-2020**.
- **CRYSP**: Gomez-Cadenas, Soleti, Lopez-Guejo, Pelegrin.
COCOA: Soleti, Gomez-Cadenas.

Team, engineering and management

- Four faculty, two engineers, two postdocs, and graduate and undergraduate students.
- Substantial experience in mechanical engineering and cryogenics.
- Strong Basque industry connections and established management: Lopez-Guejo and Pelegrin.
- Well funded by the ERC and Basque Government; additional positions planned for CRYSP and COCOA.

CRYSP and COCOA for proton beam monitoring

- The CRYSP and COCOA concepts can be applied as **high-performance, low-cost solutions** for proton beam monitoring. Our recent study shows the potential of in-room CRYSP PET.
- We have **operating prototypes of both systems** and are moving towards the construction of larger apparatus.
- The group is establishing a close collaboration with **Biogipuzkoa** and the medical staff of the forthcoming proton beam therapy centre in **Donostia**.

J.J. Gomez-Cadenas et al., *A High-Sensitivity In-Room CRYSP Scanner for Distal-Edge Verification in Proton Therapy*, paper in preparation, available upon request.

Our vision: an integrated monitoring service

- Optimised, **high-precision, low-cost apparatus**, combined with fast and efficient software using **AI-enabled technologies**, to provide real-time monitoring and support treatment correction in cooperation with doctors.
- A potential spin-off, **ARGOS**, would offer an integrated solution to proton therapy centres:

Hardware

Installation and maintenance

Software

Full development and
integration

Day-by-day service

Close cooperation with
doctors

The model could be extensively tested and optimised in **Donostia**, then expanded to new proton beam centres.

Development programme and investment

Foreseen development cycle: 4 years.

Investment: approximately €1 million per year, or €3–4 million in total.

1. Validation of prototypes.
2. Preclinical demonstrators.
3. Clinical trials and medical validation.
4. Construction of medical-grade apparatus.
5. Initial operation.

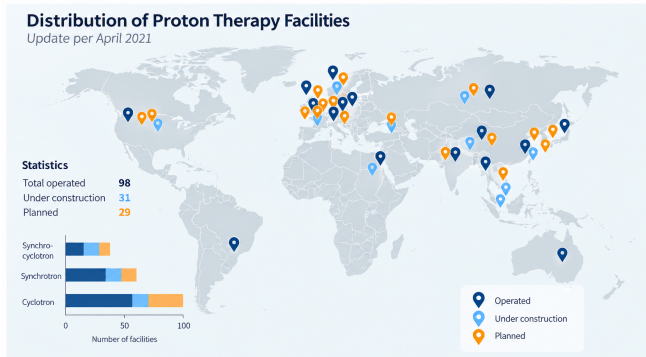
Market Size: The Vision of Independent Consultants

Source	Base year	Market size	Forecast	CAGR
Grand View Research	2023	\$749.5 M	\$1.22 B in 2030	7.3%
Knowledge Sourcing Intelligence	2025	\$959 M	\$1.27 B in 2030	5.84%
Fundamental Business Insights	2025	\$849 M	\$1.73 B in 2035	7.4%
6Wresearch	2024	\$1.30 B	\$2.40 B in 2031	8.4%
The Business Research Company	2025	\$1.43 B	\$1.58 B in 2026	10.5%
The Insight Partners	2024	\$1.56 B	\$3.27 B in 2031	11.2%
Market Glass	2024	\$696 M	\$1.50 B in 2030	13.5%

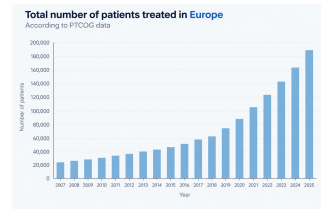
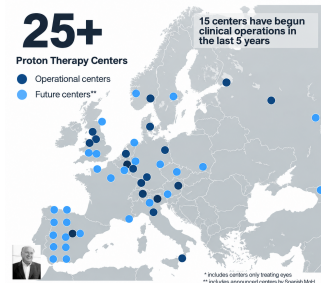
Market definitions and scopes differ across reports. Differences may reflect the inclusion or exclusion of products and services.

Independent estimates consistently place the current global proton-therapy market in the order of \$1 billion per year.

Proton therapy: worldwide deployment.



Over the last two decades, proton therapy has evolved from a highly specialised treatment into a **globally deployed clinical technology**.



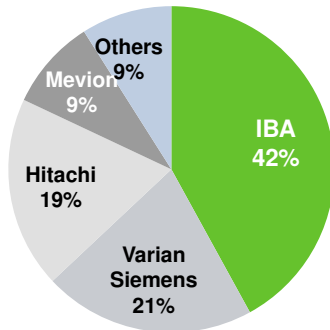
Market Share

IBA originated in Belgium in the early 1980s as a university-led effort to develop medical cyclotrons, and evolved into the world's leading provider of proton-therapy systems and particle-accelerator technology.

We will analyse several elements of information provided by this actor in the newest Financial Reports (FY2025).

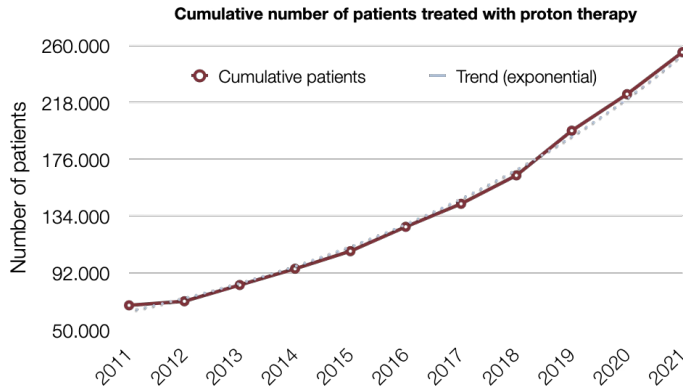
Source: IBA FY2025 Results Presentation.

Total market share in PT rooms



Research centres and Rutherford Cancer Centres excluded.// Competitors' room counts are restated by IBA to include previously unpublished or unconfirmed orders.

Proton therapy adoption



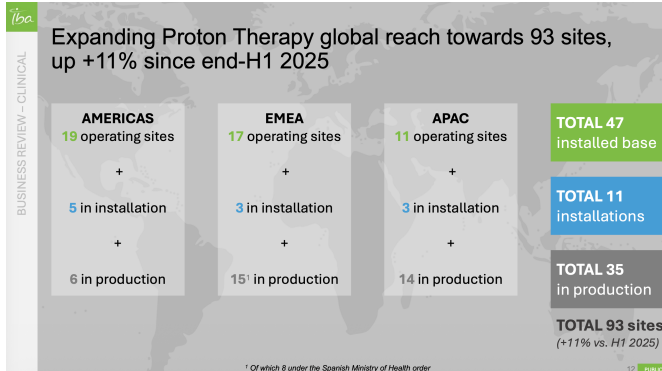
Source: PTCOG Patient Statistics.

The number of patients treated with proton therapy has increased steadily at exponential pace

- **Expanding installed base**
More centres and treatment rooms.
- **Technology maturation**
More compact and standardised proton delivery systems.
- **Growing clinical evidence**
Supporting wider adoption.

Proton therapy is becoming an established clinical technology.

Market dynamics and Prospectives



Source: IBA Half-Year Results 2025.

A market still expanding

IBA, the global market leader, reports:

45%

of IBA's current portfolio
is still to become operational

This indicates substantial
near-term growth of the installed
base.

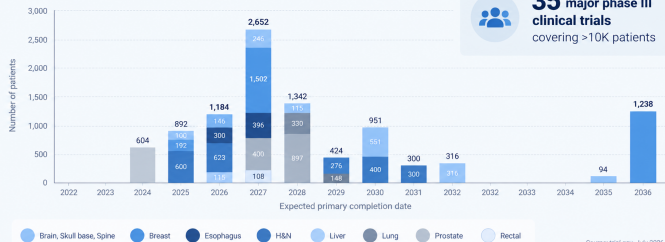
Clinical evidence approaching an inflection point

Historically, proton therapy faced an **evidence gap**: its dosimetric advantages were clear, but randomized clinical evidence of superiority over modern photon radiotherapy remained limited.

Expanding clinical evidence pipeline

Supporting momentum for proton therapy

Number of patients per ongoing phase III clinical trial (proton therapy vs. radiotherapy)



Source: IBA investor presentation, 2026; trial data from ClinicalTrials.gov.

The evidence base is changing

35+

major Phase III trials

A large concentration of trial readouts is expected between 2026 and 2030.

Early Phase III results:

- **Oropharyngeal cancer:** IMPT showed non-inferior disease control and lower severe toxicity than IMRT. Frank et al., *The Lancet*, 2025.
- **TORPEdO** IMPT and IMRT showed similar quality of life and disease control.

The Proton Therapy Paradox

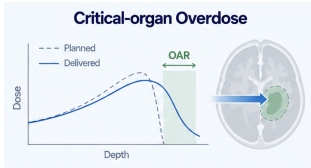
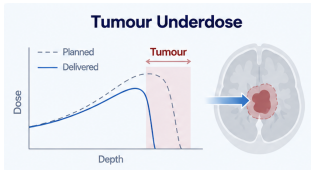
Sources of uncertainty

- **Anatomical changes:** Tumour and organ variation
- **Positioning & motion:** Setup and breathing
- **Imaging & modelling:** Stopping-power and range estimation
- **Beam delivery:** Energy, range and spot position



Uncertainty in where the beam actually stops

Clinical consequence



Conservative clinical response

- **Robust treatment planning:** Range setup uncertainties are explicitly incorporated into the plan.
- **Conservative beam selection:** Avoidance of beam paths where range errors could place the distal edge inside a critical structure.
- **Loss of conformality:** Extra robustness can increase irradiation of normal tissue and partially erode the physical advantage of protons.
- **More verification and adaptation:** Repeat imaging, robustness evaluation and, when needed, adaptive replanning.



Uncertainty can restrict optimal beam arrangements and reduce the expected benefit in anatomically variable or particularly demanding cases.

Range uncertainty is a clinically relevant limitation

Range uncertainty has measurable dosimetric consequences and already drives additional clinical adaptation workflows.

Dosimetric impact

Tarp et al., 2024 Reducing assumed range uncertainty from 3.5% to 2.0% reduced irradiation of normal tissue in brain, head-and-neck and breast plans. *Phys. Imaging Radiat. Oncol.* 31 (2024) 100632

Taasti et al., 2023 Reducing range uncertainty from 3% to 2% reduced dose to at least one OAR in **89% of patients**; **44%** showed a predicted reduction in toxicity. Clinical benefit of range-uncertainty reduction

EPTN systematic review, 2024 Higher proton dose conformity comes at the cost of greater sensitivity to physical, anatomical and positioning uncertainties. *Radiother. Oncol.* 197 (2024) 110365

Clinical workflow impact

Trnkova et al., 2023 **84% of surveyed particle-therapy centres** use treatment adaptation to account for changes occurring between treatment fractions. A major barrier was the lack of integrated and efficient workflows. *Phys. Imaging Radiat. Oncol.* 26 (2023) 100442

Clinical management review, 2024 Range errors can cause target undershoot or distal overshoot into normal tissue, compromising part of the theoretical dosimetric advantage of particle therapy. Beam range uncertainty and its clinical management

Reaction from the Industry

The major players already recognise the significance of the issue and are adapting their strategies accordingly.

		IBA Clinical strategic progress			
BUSINESS REVIEW – CLINICAL		Proton Therapy		Dosimetry	
		Presented at CMD	Progress	Presented at CMD	Progress
1	Increase our technology penetration in core markets	- Increased access with clinical evidence - Improved relevance with product innovation	ConformalFLASH® FDA IDE approval for first-in-human trial Launch of AdaPTInsight XR imaging upgrade China NMPA trials: all primary endpoints met	Value capture through unique combined portfolio of Imaging and RT QA products	myQA StarTrack® released at ESTRO 2026 Publication of preprint: on phantoms for imaging and AI QA
2	Expand in high potential geographies	Asia, esp. China: accelerated development	Strong commercial traction in Europe and the Americas, with 5 rooms sold Installations advancing in Spain and China	US: leadership growth	US: continued regional-specific challenges
3	Expand along the value chain	- Improved service profitability thanks to scale-up and efficiency initiatives - Expanded modalities coverage across heavy particles	PT turnaround confirmed: Adj. EBIT €12.3M, 7.3% margin Services revenues up 3%, 47 active sites Improvements in operational efficiency	Greater share of QA value chain through acquisitions and partnerships	Backlog towards more Proton Therapy QA driven, with revenue generation backloaded

In-vivo verification: towards intervention

The next step is to turn patient-specific verification into a decision: **does this fraction require additional attention?**

What should trigger a review?

- Unexpected range shifts.
- Changes consistent with altered anatomy.
- Possible setup or modelling errors.

Towards an intervention workflow

1. **Detect** a deviation from expectation.
2. **Review** its cause and clinical relevance.
3. **Act**: additional imaging or replanning when warranted.

A detected deviation is a trigger for review; a validated workflow must determine when intervention is needed.

References

Mogliani et al., *Front. Oncol.* 12 (2022) 929949 — in-beam PET and inter-fractional changes.
Xie et al., *Int. J. Radiat. Oncol. Biol. Phys.* 99 (2017) — clinical prompt-gamma verification.
Bertschi et al., *Med. Phys.* 52 (2025) e70142 — verification towards adaptive workflows.

Why has in-vivo verification not become routine?

Clinical feasibility is established; routine use requires a reliable, actionable result within the treatment workflow.

1. Reliable measurement

Precision depends on signal statistics, background, anatomy and detector geometry.

Need: quantified uncertainty at clinical beam rates.

2. Clinical interpretation

A range shift alone does not establish the effect on target coverage or healthy tissue.

Need: validated action thresholds linked to dose impact.

3. Workflow integration

Positioning, calibration, analysis and review all compete with treatment time.

Need: automated checks, rapid feedback and a defined response.

4. Evidence and value

Detecting a deviation is only one part of demonstrating a useful clinical service.

Need: evidence that decisions improve care and justify time and cost.

Adoption requires the full chain: reliable measurement, justified action and practical delivery.

Selected evidence Hueso-González et al., PMB 63 (2018) 185019: clinical-scale prototype.

Richter et al., Radiother. Oncol. 118 (2016): first clinical PGI application. Bertschi et al., PhiRO 34 (2025) 100778: adaptive-workflow feasibility.

Two decades of in-vivo verification: what have we learned?

System	Clinical achievement	Strength	Remaining gap
BOLPs-RGp ; Japan, 2010	Gantry-mounted PET used in 48 patients .	Daily visualization of changes in the proton-irradiated volume.	Open PET geometry; indirect activity–range relationship; post-irradiation acquisition.
NeuroPET ; MGH, 2011	Mobile full-ring PET installed inside the treatment room.	Acquisition started only a few minutes after irradiation without patient repositioning.	Still post-treatment; biological washout and additional workflow remain.
Prompt Gamma ; IBA / OncoRay / Penn, 2016–2021	First clinical PGI; later clinical PBS verification and ~ 1 mm range-validation uncertainty.	Measurement occurs during beam delivery .	Dedicated hardware and interpretation workflow; clinical use remained study/prototype based.
INSIDE ; CNAO, 2019–present	In-beam PET integrated in a clinical treatment room; detected anomalous inter-fractional changes.	Direct integration with clinical hadron-therapy.	Limited-angle PET geometry and complex conversion from activity changes to actionable clinical decisions.
Range probing / PGI ; 2020–2025	Clinical range probing and increasingly fast PGI are being developed as safeguards for adaptive therapy.	Designed explicitly to detect fractions requiring additional attention.	The remaining step is an validated intervention workflow : detect → review → re-image/replan.

Repeated clinical feasibility has not yet produced a universal workflow that converts a detected deviation into an immediate clinical action.

Main Challenges Ahead

Development priorities: demonstrate that the system can fit the room, produce trustworthy results and support clinical action.

1. Integrate the hardware

Balance detector coverage with patient access, couch motion and beam clearance.

Milestone: demonstrate operation in a representative treatment geometry.

3. Test the response workflow

Define who reviews an alert, what confirms it and when to re-image or replan.

Milestone: measure missed deviations, false alerts and decision time.

2. Validate the information

Quantify sensitivity to range shifts, background, anatomy and model uncertainty.

Milestone: benchmark against independent imaging and dose calculations.

4. Build the adoption case

Co-develop the intended use, regulatory strategy and service model with partners.

Milestone: quantify clinical value, staff workload and total ownership cost.

Success means a validated clinical service, with hardware, software and response protocol.

Proposed development milestones. Technical and clinical precedents:

Hueso-González et al., PMB 63 (2018) 185019; Richter et al., Radiother. Oncol. 118 (2016); Bertschi et al., PhiRO 34 (2025) 100778.

Strategy to overcome the translation barriers

Develop the technology, clinical use case and regulatory pathway in parallel — not sequentially.

1. Build on a common detector platform

CRYSP can support both **PET** and **Compton-camera** architectures, allowing the technology to address complementary in-vivo verification approaches.

The detector concept is based on **non-strategic materials**, avoiding rare-earth supply dependence, while retaining a **competitive detector cost**.

3. Design for certification from day one

Implement an **ISO 13485-compliant quality management system** from the beginning of development.

Integrate design controls, traceability, risk management, verification and validation into the R&D process rather than retrofitting them after the prototype stage.

Objective: reduce redesign and shorten the path from research prototype to medical device.

2. Co-develop the clinical use case

Work with **BioGipuzkoa** and the clinical teams at **Onkologikoa** from the development phase.

Use the new proton-therapy facility, expected to enter operation in **2027**, to define realistic treatment geometries, validation protocols and prospective clinical studies.

Key question: what measured deviation should trigger review, additional imaging or replanning?

4. Validate towards clinical action

Progress through a staged validation programme:

detector performance → **beam/phantom validation** → **treatment-room integration** → **clinical validation**

Define quantitative go/no-go criteria for sensitivity, false alerts, acquisition time and decision relevance.

The endpoint is not simply a better image, but a **trustworthy clinical decision tool**.

Potential Strategic Buyers and Exit Routes

A successful in-vivo verification platform could be strategically valuable to companies controlling proton delivery, radiotherapy workflow or treatment QA.

Proton-therapy OEMs

IBA

Global PT market leader, with a large installed base, service business and direct interest in treatment verification.

Siemens Healthineers / Varian

Strong fit with integrated imaging, treatment delivery, adaptive radiotherapy and precision-therapy workflows.

Hitachi

Major global proton-therapy supplier with direct access to treatment systems and clinical sites.

Strategic rationale

Integrate in-vivo verification directly into the treatment platform and differentiate the next generation of proton-therapy systems.

QA and dosimetry players³

Sun Nuclear / Mirion

Established radiotherapy QA portfolio and demonstrated interest in expanding towards digital treatment QA.

PTW

Strong position in radiation dosimetry and patient/machine QA; in-vivo verification is a natural adjacent market.

Specialised QA vendors

Potential buyers if the product develops as an independent verification layer rather than an accelerator subsystem.

Strategic rationale

Extend QA from verification before treatment to verification of what was actually delivered.

Workflow and software players

RaySearch Laboratories

Relevant if measured range information becomes integrated with adaptive planning and clinical decision support.

Broader oncology-platform companies

Potential interest increases if the project evolves from detector hardware into a verification and decision-support platform.

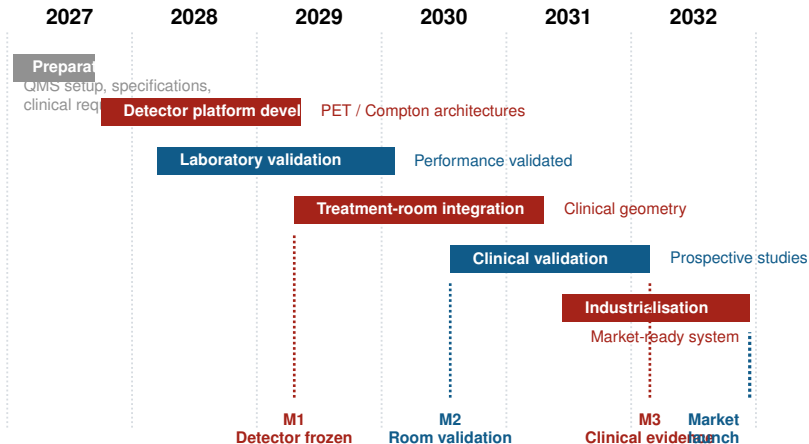
Most likely exit route

Strategic acquisition by a proton-therapy OEM or a radiotherapy QA company after clinical validation and early market deployment.

The strategic value is not only the detector: it is the ability to connect beam delivery, verification and clinical action.

Development Roadmap

Illustrative roadmap — activities and milestones to be refined



Founding Team

Complementary expertise from detector R&D to industrialisation.

Juan José Gómez-Cadenas

Scientific leadership

Ikerbasque Research Professor at DIPC with an international career at Stanford/SLAC, CERN, Harvard, UMass, CSIC and the University of Valencia. An expert in particle physics and nuclear instrumentation, he leads NEXT and has received ERC Advanced and Synergy Grants. He brings extensive experience in detector development and medical-imaging technologies, including PETALO and CRYSP.

Francisco López Gejo

Industrialisation & Business Development

Physicist whose career has combined academic research and industry, with experience in medical technology development, R&D management and entrepreneurship. He was co-founder and General Manager of BihurCrystal, a Deep Tech venture focused on transferring technologies from fundamental research to industry. At DIPC, he works at the interface between research, technology transfer and project development.

Vicente Herrero Bosch

Electronics, DAQ & System Integration

Researcher at the Institute for Instrumentation in Molecular Imaging (I3M, CSIC-UPV), specialising in front-end electronics, SiPM readout and high-performance DAQ for particle detectors and molecular imaging. He has contributed to NEXT, PETALO and CRYSP, bringing expertise in electronics, data acquisition and scalable detector integration.

Stefano Roberto Soleti

Detector Technology & Experimental Development

Ikerbasque Research Fellow at DIPC and PhD in Particle Physics from the University of Oxford, with previous experience at Berkeley Lab, Harvard, INFN Frascati and Fermilab. Co-author of the CRYSP and COCOA concepts, his work focuses on cryogenic scintillators, SiPM readout, monolithic PET detectors and compact Compton cameras.

Funding Strategy

Our funding strategy aims to leverage public funding wherever possible to cover a significant share of the technical and clinical de-risking, thereby reducing the private capital required before market entry.

1. Functional prototypes — already funded

Basque Government ESS funding — already allocated to DIPC.

€ 1.0 M over 2027–2028 to develop functional prototypes of both the **Compton camera** and the **PET scanner**.

2. Validation in a relevant environment

ELKARTEK 2027 — collaborative project with **BioGipuzkoa**. Currently under preparation

Target budget: € 1.5 M over two years for validation of the detector technologies in a clinically relevant environment.

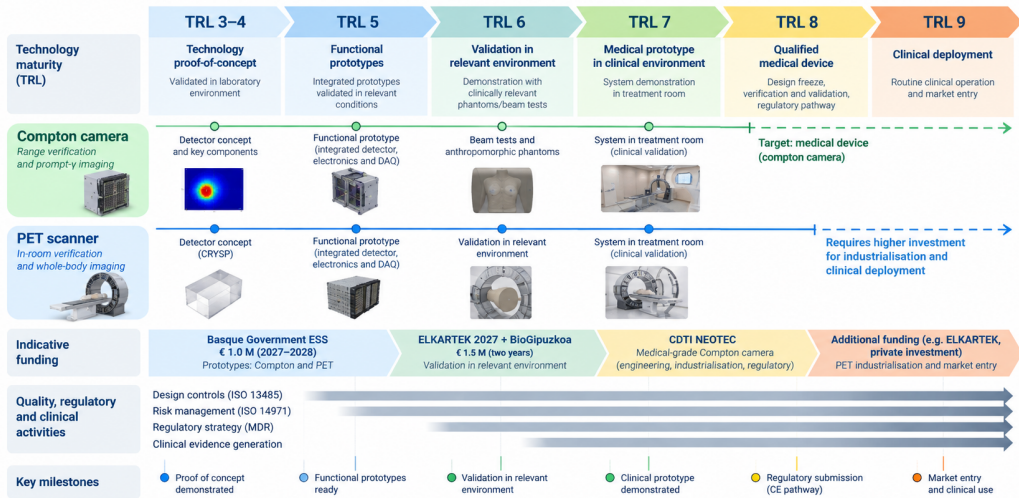
3. Medical-device development

CDTI NEOTEC — proposed funding route for development of a **medical-grade Compton camera**, supporting engineering, industrialisation and regulatory preparation after company incorporation.

4. PET industrialisation requires a larger investment

The PET system requires substantially higher capital investment than the Compton-camera programme. Its development could therefore combine **private**

Development Roadmap



VISION

Make in-vivo verification a routine component of proton therapy.

Treatment decisions should rely not only on what was planned, but also on evidence of what was actually delivered.

MISSION

Develop affordable, clinically deployable PET and Compton systems for proton-therapy verification.

Built around a common detector platform and designed from the outset for clinical integration and medical-device translation.

Installed-system economics

PET + Compton camera + software: a complete, commissioned monitoring system.

Estimated cost to deliver one system

Component / activity	€ k
PET ring, BGO monolithic, 600 mm bore, 300 mm length	740
Cryogenic system, -80°C	160
Compton camera	100
Total raw hardware budget	1,000
Assembly, integration, computing, installation, validation and warranty allowance	350
Total delivered cost	1,350
First service year included	100
Total cost per installed system	1,450

10 installations represent € 25 M of system revenue.

Backup: CsI(Tl) ring in the same geometry, operated at room temperature, € 600 k and no cryogenics, € 1,050 k delivered. CsI(Tl) delivers acceptable performance at lower cost and with no dependence on a restricted material. BGO delivers the best performance.

Installed system price

€ 2.5 M

per installed system, excluding VAT; first service year included

€ 1.1 M gross profit
42.0% gross margin

Sensitivity: at € 1.6 M delivered, 32% margin;
at € 1.9 M, 20%.

Bottom-up installation market

Scenario at € 2.5 M per installed system: new rooms and retrofits.

New-room opportunity

IBA: 12 room orders in 2025, at 63% share.
Approximately 19 global room orders/year.

At 100% monitoring uptake: 19 systems/year.

€ 47.5 M/year

Retrofit scenario

Assume 300 existing treatment rooms could accommodate ARGOS.

At 5% adoption per year:
15 installations at € 2.5 M each.

€ 37.5 M/year

19 new-room installations + 15 retrofits

€ 85 M/year in potential system sales

Recurring services are a separate market, calculated on active paying contracts.

Illustrative scenario; room compatibility and adoption require validation.

Source: IBA FY2025 results: room orders and share.

Annual services: scope and economics

Per fraction: online Compton red-light check, in-room PET within minutes, and AI analysis giving a go/no-go for the next fraction, the distal edge, fraction statistics and updated coverage. Data and models stay on local machines. An ARGOS specialist on site interprets results with the treating team, by tumour type.

Annual direct cost and scope	€ k
Hardware maintenance	30
Cryocooler, spares, calibration and service travel	
Dedicated ARGOS specialist + cover	80
Daily QA, per-fraction analysis and interpretation with the treating team	
Local software operating expenses	5
Liability insurance and post-market surveillance	15
Total annual direct cost	130

Annual service fee

€ 400 k

per installation, excluding VAT; 5-year contract after the included first year

€ 270 k gross profit
67.5% gross margin

Central software engineers and company overhead are funded from service gross profit.

First service year included in the system price; the paid contract covers the following 5 years, includes training of hospital personnel and ends with handover after 6 years on site. Thereafter a € 50 k/year hardware fee (€ 20 k gross profit).

Personnel costs are estimated for Spain, where the project will be funded: baseline salary € 50 k + 32% employer costs, 20% relief allowance, € 79.2 k rounded to € 80 k.

Recurring services: annual revenue potential

Each active annual contract generates € 400 k revenue and € 270 k gross profit.

Active contracts	Revenue € M/year	Direct cost € M/year	Gross profit € M/year
10	4.00	1.30	2.70
100	40.00	13.00	27.00
300	120.00	39.00	81.00

Recurring revenue grows with the serviced installed base, independently of new-system sales.

Each paid contract covers 5 years after the service year included with the system; after handover an installation pays the € 50 k/year hardware fee.

Proton therapy: market growth

Treatment rooms worldwide: provisional starting base of 300 in 2026.

	Assumed annual growth	2036	2046
Slow	3%	403	542
Central	5%	489	796
Fast	7%	590	1,161

Current evidence

PTCOG: 128 operating proton centres (January 2026); about 57 proton rooms listed under construction (June).

IBA, first half 2026: 5 room orders at 31% share imply about 16 global orders; 46 IBA equipment projects in backlog.

Yap et al., Front. Oncol. 16 (2026) 1791102: operating centres. PTCOG, June 2026: construction list; room tally excludes carbon-only rooms. IBA H1 2026 presentation, 27 August, pp. 12–13: sites and room orders. IBA H1 2026 results, p. 4: equipment backlog.

Total potential market

All suppliers combined, at full room compatibility and adoption.

€ 2.5 M per system with the first service year included; € 400 k per year for the following 5 years, then € 50 k per year.

Central growth case	2036	2046
Operating treatment rooms	489	796
Net additional rooms since 2026	189	496
Cumulative new-system potential (€ M)	472	1,240
Annual service potential (€ M/year)	59	95

Systems: cumulative sales associated with net room expansion; excludes existing-room retrofits and replacements.

Services: rooms in their included year pay nothing, rooms in the following 5 years pay the full fee, older rooms the hardware fee; full-year contracts.

Funding request

Use of funds

	€ k
Hardware and prototype	1,600
Personnel	1,730
Clinical, regulatory and operations	315
Programme reserve	252
Backup	103
Total	4,000

€ 4 M

Over 4 years, 2029–2032

PET + Compton camera

Integrated software

Clinical monitoring service

Development and validation with
Onkologikoa and BioGipuzkoa.

From phantom tests to a complete clinical demonstrator.

Private round after € 2.5 M of public funding completes technology development and first pre-clinical measurements in 2027–2028.

Operations: hospital programme € 50k; regulatory work € 25k; premises € 40k; IT € 80k; travel € 120k.

Development team

Personnel budget: € 1.73 M. Annual figures include employer costs.

Role	FTE	€ k/year	Years	Total € k
CEO / project lead	1	100	4	400
Detector and electronics engineer	1	85	4	340
Software engineer	1	85	4	340
Medical physicist / validation lead	1	85	4	340
Assembly and laboratory technician	1	50	3	150
Administrative clerk	1	40	4	160
Total				1,730

Hardware integration, reconstruction software, phantom measurements and clinical validation.

Building the full demonstrator

1 PET + 1 Compton camera, integrated software and monitoring service.

Hardware and prototype

	€ k
PET materials	740
Compton camera	100
Cryogenic system	160
Computing and other hardware	100
Fabrication and test equipment	200
First-device contingency	300
Total	1,600

Hospital programme: € 50k. External regulatory and conformity work: € 25k.

Development with Onkologikoa and BioGipuzkoa

Year 1 (2029) — Medical-grade design

Design controls, quality system, software architecture and clinical protocol.

Year 2 (2030) — System build and phantom tests

Assemble the system; test accuracy, reliability and service procedures.

Years 3–4 (2031–2032) — Clinical validation

Clinical investigations, workflow validation and conformity assessment.